

Q 2 ii) $a = 1 \text{ cm}$ $b = 3 \text{ cm}$

$I = 60 \text{ A}$

Find $\vec{H}$ at $r = 2 \text{ cm}$

Ampere's Law

$$\oint \vec{H} \cdot d\vec{l} = I_{\text{enclosed}} = H \cdot (2\pi r)$$

$$\therefore H = \frac{I_{\text{enclosed}}}{2\pi r}$$

— (2)

a) $r = 2 \text{ cm}$

$$I_{\text{enc}} = J \cdot A_{\text{enc}}$$

— (1)

$$J = \frac{I}{\pi(b^2 - a^2)} = \frac{60}{\pi(0.03^2 - 0.01^2)} = \frac{75000}{\pi} \text{ A/m}^2$$

— (2)

$$A_{\text{enc}} = \pi(r^2 - a^2) = 3\pi \times 10^{-4}$$

— (2)

$$\therefore I_{\text{enc}} = 22.5 \text{ A}$$

$$\therefore \vec{H} = \frac{22.5}{2\pi \cdot (0.02)} = 179.1 \text{ A/m}$$

— (1)

b) $r = 5 \text{ cm}$

$$H = \frac{60}{2\pi(0.05)} = 191.1 \text{ A/m}$$

— (2)

Q 3. ii) $A = 0.01 \text{ m}^2$
 $d = 0.002 \text{ m}$

$$\vec{E}(t) = 10 \cos(10^6 t) \text{ V/m}$$

a) $J_d = \epsilon_0 \frac{\partial \vec{E}(t)}{\partial t}$ — (4)

$$J_d = -8.85 \times 10^5 \sin(10^6 t) \text{ A/m}^2$$

b) $I_d(t) = J_d \cdot A$ — (3)

$$= J_d \cdot (0.01)$$

$$I_d(t) = -8.85 \times 10^{-7} \sin(10^6 t) \text{ A}$$

c) Comparison at three points. — (3)

Q 4. $f = 10 \text{ GHz}$, $\sigma = 5.8 \times 10^7 \text{ S/m}$, $\mu_r = 1$

a) Skin depth $\delta = \sqrt{\frac{2}{\omega \mu \sigma}}$ — (3)

$$\omega = 2\pi f$$

$$\boxed{\delta = 0.209 \mu\text{m}}$$

b) $\alpha = \frac{1}{\delta} = 4.78 \times 10^6 \text{ Np/m}$ — (2)

c) Significance of skin depth — (5)

Q1 ii) $\vec{E} = 4x \vec{a}_x + 2y \vec{a}_y \text{ V/m}$

$$V_{AB} = V_A - V_B = - \int_B^A \vec{E} \cdot d\vec{l}$$

$$\int_{(0,0)}^{(2,0)} \vec{E} \cdot d\vec{l} = \int_0^2 4x \, dx = 8 \quad \text{--- (2)}$$

$$\int_{(2,0)}^{(2,2)} \vec{E} \cdot d\vec{l} = \int_0^2 2y \, dy = 4 \quad \text{--- (2)}$$

$$\int_B^A \vec{E} \cdot d\vec{l} = 8 + 4 = 12$$

$$\therefore \boxed{V_{AB} = -12 \text{ V}}$$

--- (1)

iii) $q_1 = 3 \mu\text{C}$ at $(0,0)$

$q_2 = -2 \mu\text{C}$ at $(3,0)$

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \cdot \vec{r}$$

--- (1)

Force on q_1 due to q_2

$$\vec{F}_{12} = 6 \times 10^{-3} \vec{a}_x \text{ N}$$

(Force along x -axis only)

$$\vec{F}_{21} = -6 \times 10^{-3} \vec{a}_x \text{ N}$$

--- (1)



Semester: January 2025 – April 2025

Maximum Marks: 50		Examination: End-Semester Examination		Duration: 2 Hrs.	
Programme code: 03			Class: SY		Semester: IV (SYU 2023)
Programme: B.Tech EXTC					
Institute/School/Department: K. J. Somaiya School of Engineering			Name of the department/Section/Center: EXTC		
Course Code: 216U03C405			Name of the Course: Electromagnetic Field Theory		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary					

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	
i)	Explain Poisson's and Laplace's equation. State their significance and give physical examples of where each is applied in electrostatics. <i>eqn — (2) explanation — (3)</i>	05
ii)	The electric field in a region is given as: $E = 4x a_x + 2y a_y$ V/m Find the potential difference between point A(2, 2) and point B(0, 0)	05
iii)	Two-point charges are placed as follows: $q_1 = +3 \mu C$ at (0, 0) $q_2 = -2 \mu C$ at (3 m, 0) Find the force on each charge.	05
Q.2	Attempt any one <i>(2) → (5)</i>	10
i)	What are magnetic boundary conditions? Derive the boundary conditions for magnetic fields at the interface of two magnetic media with different permeabilities. Support your answer with appropriate diagrams. <i>→ (3)</i>	
ii)	Two coaxial cylindrical shells, inner radius $a=1$ cm, outer radius $b=3$ cm, carry a total current of 60 A uniformly distributed in the annular region. Using Ampere's Law, calculate the magnetic field intensity H: (a) At $r=2$ cm (b) At $r=5$ cm	
Q.3	Attempt any one <i>→ (6)</i>	10
i)	Compare and contrast Transformer EMF and Motional EMF with suitable mathematical derivations. Explain their applications in devices such as transformers, electric generators. <i>→ (4)</i>	
ii)	A parallel-plate capacitor with a plate area of 0.01 m^2 and a separation of 2 mm is connected to an AC source producing a time-varying electric field given by: $E(t)=10 \cos(10^6 t) \text{ V/m}$ 1. Calculate the displacement current density between the plates. 2. Find the total displacement current if the capacitor is filled with air ($\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$). 3. Compare the displacement current with the conduction current for different materials (dielectric vs. conductor).	